MARYAM REHAN

INTERNSHIP REPORT

CYBER SECURITY (SOC)

Week 5 Task 13 — Phishing Email Analysis

Objective

To investigate a suspicious phishing email by analyzing its headers, checking sender authentication (SPF, DKIM, and Return-Path), identifying sender spoofing, and recognizing phishing indicators.

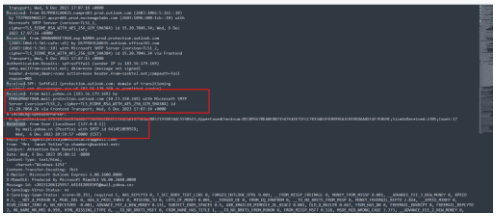
Explanation

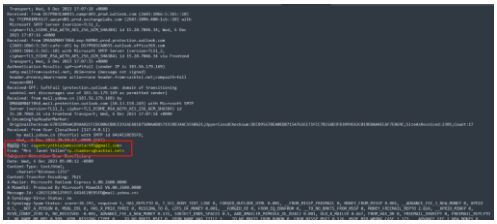
I analyzed a sample email titled “Attention Dear Beneficiary.” The email claimed that the recipient had been selected to receive $16,000,000 in COVID relief funds. I examined the From, Return-Path, and Reply-To fields and found that the Reply-To address did not match the claimed sender domain. The SPF result was softfail, while DKIM and DMARC were not present in the available analysis. The originating server and sender IP were also reviewed. The email contained no URL or attachment, so a VirusTotal URL scan was not applicable to this sample.

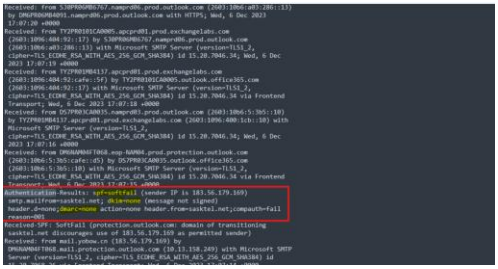
Conclusion

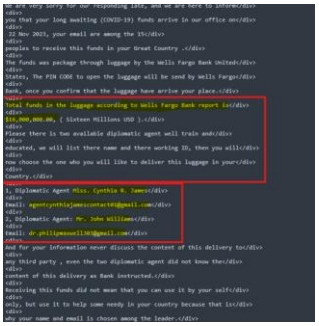
The email was identified as a malicious advance-fee phishing scam. This task helped me understand email header analysis, sender spoofing, authentication failures, and common phishing red flags used in social engineering attacks.

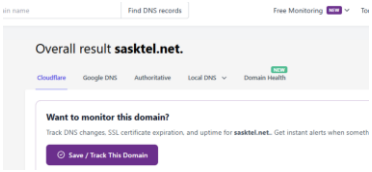
Screen Shoot

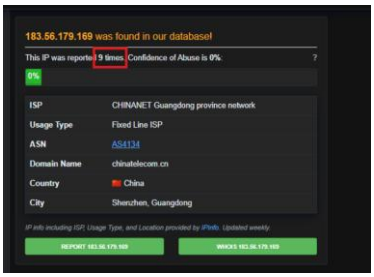




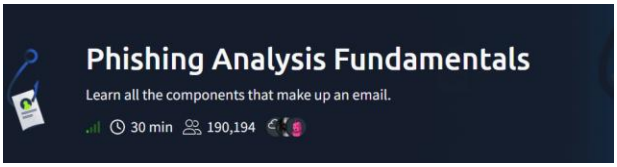

Authentication-Results: spf=softfail
Received-SPF: SoftFail


Total funds in the luggage according to Wells Fargo Bank report is


Overall result sasktel.net.
Cloudflare
Google DNS
Authoritative
Local DNS
Domain Health
Want to monitor this domain?
Save / Track This Domain


183.56.179.169 was found in our database!
This IP was reported 9 times. Confidence of Abuse is 0%
0%
ISP
CHINANET Guangdong province network
Usage Type
Fixed Line ISP
ASN
AS4134
Domain Name
chinatelecom.cn
Country
China
City
Shenzhen, Guangdong

TRY HACK ME


Phishing Analysis Fundamentals
Learn all the components that make up an email.
30 min
190,194

Task 1 Introduction

Task 2 The Email Address

Task 3 Email Delivery

Task 4 Email Headers

Task 5 Email Body

Task 6 Types of Phishing

Task 7 Conclusion

Answer the questions below

I am ready to learn about phishing analysis!

No answer needed ✓ Correct Answ

Answer the questions below

Identify the domain used in the following email address:
`hatsalesman@tryhatme.com`

tryhatme.com ✓ Correct Answer

Answer the questions below

Which protocol is responsible for sending an email from a client to a mail server?

SMTP ✓ Correct Answer

Which service is used to look up the recipient domain's mail server?

DNS ✓ Correct Answer

Bob wants to access his email from multiple devices, including his phone and laptop.
Which protocol should he use?

IMAP ✓ Correct Answer

Answer the questions below

What is the full subject line of `email1.eml`?

Help protect your budget by protecting your home ✓ Correct Answer

View the message source of `email1.eml` using Thunderbird in your VM.
What the IP address listed as the `X-Originating-Ip`?

43.255.56.161 ✓ Correct Answer

Answer the questions below

Open up the `email2.txt` file to view the source of an attachment.
What is the `Content-Type` of the attachment?

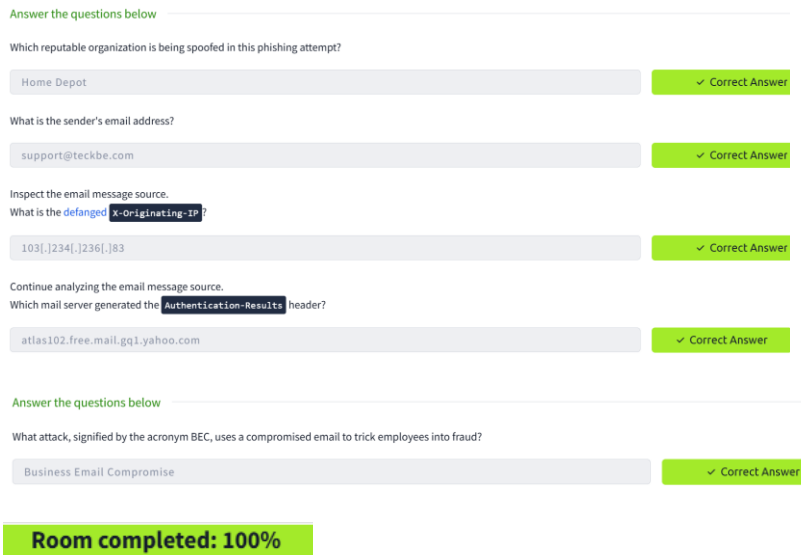
application/pdf ✓ Correct Answer

What is the name of the attachment from the previous question?

zmqpalgh.pdf ✓ Correct Answer

Decode the base64 string using either a PDF converter or CyberChef.
What is the hidden flag value?

THM{BENIGN_PDF_ATTACHMENT} ✓ Correct Answer



Room completed: 100%

Objective

To understand the fundamentals of phishing email analysis, learn how emails are delivered, examine email headers and body content, and identify different types of phishing attacks.

Explanation

I completed the TryHackMe Phishing Emails room, covering email address structure, email delivery protocols (SMTP, POP3, and IMAP), and email header analysis. I learned how to identify sender information, Reply-To mismatches, and authentication details. I also studied email body inspection, attachment analysis, common phishing techniques, and Business Email Compromise (BEC). The practical exercises helped me recognize suspicious email indicators and understand safe email investigation.

Conclusion

This task improved my understanding of phishing detection and email investigation. I learned how to identify suspicious sender addresses, analyze email headers, recognize malicious attachments, and understand social engineering techniques used by attackers. These skills are useful for performing basic phishing investigations as a SOC L1 Analyst.